10.19 One Self-Energy, Three Regulators

LecNote12 computes the one-loop self-energy of φ^3 theory in $d = 6$ with dimensional regularization, and sketches two other routes on the way: a red "Skip" block on pp. 6–7 that regulates nothing, and Pauli–Villars on p. 8. Students ask:

- "What is the trick in the Skip block, and do I need it?"
- "What does the Pauli–Villars Π look like, and what cancels its Λ's?"
- "p. 6 rotates the contour only if the integrand falls faster than $1/\bar{q}^{\,d}$; in $d = 6$ the bubble does not. How can dim reg rotate it?"
- "Why do three methods give the same Π? What does each hide?"

In short: the divergences of the bubble sit in a polynomial $a + bk^2$, the form of the counterterms $Ak^2 + Bm^2$. Once A and B are fixed by $\Pi(-m^2) = \Pi'(-m^2) = 0$, the regulator has nothing left to decide, and all three routes give p. 13's Π. They differ in the counterterms: Pauli–Villars puts $\ln\Lambda$ and Λ^2 into them, dim reg $1/\epsilon$ and no Λ^2, and the regulator-free route never writes them down.

Conventions are Srednicki's and LecNote12's: metric $(-,+,+,+)$, $k^2 = -m^2$ on shell, counterterms $-\frac{1}{2}A\,\partial\varphi\partial\varphi - \frac{1}{2}Bm^2\varphi^2$, $\alpha = g^2/(4\pi)^3$. After Feynman parameters and $d^dq = i\,d^d\bar{q}$ (pp. 4–6),

$$\Pi(k^2) = \tfrac{1}{2}g^2 I(k^2) - Ak^2 - Bm^2, \qquad I(k^2) = \int_0^1 dx \int \frac{d^d\bar{q}}{(2\pi)^d}\,\frac{1}{(\bar{q}^2 + D)^2}, \tag{10.19.1}$$

with $D = x(1-x)k^2 + m^2$ and $D_0 \equiv D|_{k^2=-m^2} = [1 - x(1-x)]m^2$. The target is p. 13's on-shell self-energy, which can also be put in p. 12's form (to $O(\alpha^2)$):

$$\Pi(k^2) = \frac{\alpha}{2}\int_0^1 dx\, D \ln\frac{D}{D_0} - \frac{\alpha}{12}(k^2 + m^2) \tag{10.19.2}$$

$$= \frac{\alpha}{2}\int_0^1 dx\, D \ln\frac{D}{m^2} + \alpha\Big(\frac{\kappa_A}{6}k^2 + \kappa_B m^2\Big). \tag{10.19.3}$$

Since $\int D\ln(D_0/m^2)\,dx$ is linear in k^2, the second line follows from the first, with $\int_0^1 x(1-x)\ln(1-x+x^2)\,dx = \frac{\pi\sqrt{3}}{6} - \frac{17}{18}$ and $\int_0^1 \ln(1-x+x^2)\,dx = \frac{\pi}{\sqrt{3}} - 2$. So the constants p. 12 leaves open are $\kappa_A = \frac{7}{3} - \frac{\pi\sqrt{3}}{2} \approx -0.387$ and $\kappa_B = \frac{11}{12} - \frac{\pi\sqrt{3}}{6} \approx 0.0098$, and $\Pi(0) = \alpha\kappa_B m^2$. The two on-shell conditions are independent linear equations for them, so the answer to p. 14's "is the assignment unique?" is yes.

Route 1: no regulator (the Skip block)

Under the integral, $\partial_{k^2}^n(\bar{q}^2+D)^{-2} = (n+1)!\,[-x(1-x)]^n(\bar{q}^2+D)^{-2-n}$: each k^2-derivative brings down $-x(1-x)$ and one more propagator power. So $I^{(n)}$ has superficial degree $d-4-2n$ and diverges for $d \geq 4+2n$ (p. 7). In $d=6$, I and I' diverge while I'', I''', … converge: the divergent part of Π is at most linear in k^2. That is why the renormalized Π cannot depend on the regulator. (Two counterterms suffice for this diagram while I'' converges, $d<8$: p. 7's bound. It says nothing about higher orders; φ^3 is non-renormalizable for every $d>6$, §10.14, §10.20.)

The Skip block uses this directly: two derivatives remove A and B, and $\Pi'' = \frac{1}{2}g^2 I''$ needs no regulator. The master formula (10.18.1) with $n=4$, $d=6$ gives $\int d^6\bar{q}\,(2\pi)^{-6}(\bar{q}^2+D)^{-4} = 1/[6(4\pi)^3 D]$, so

$$\Pi''(k^2) = \frac{\alpha}{2}\int_0^1 dx\,\frac{x^2(1-x)^2}{D}. \tag{10.19.4}$$

Then "$\int dk^2$ twice" (p. 6): Taylor's formula with integral remainder,

$$\Pi(k^2) = \Pi(-m^2) + \Pi'(-m^2)\,(k^2+m^2) + \int_{-m^2}^{k^2} dt\,(k^2-t)\,\Pi''(t), \tag{10.19.5}$$

puts the on-shell conditions where the integration constants go, and the first two terms vanish. In the third, D is linear in t with slope $x(1-x)$; $u = D(t)$ turns it into $\frac{\alpha}{2}\int_0^1 dx \int_{D_0}^{D} du\,(D-u)/u$, with $\int_{D_0}^{D} du\,(D-u)/u = D\ln(D/D_0) - (D-D_0)$ and $\int_0^1 (D-D_0)\,dx = \frac{1}{6}(k^2+m^2)$. That is (10.19.2), with no ϵ, no Λ and no counterterm in sight.

Route 2: Pauli–Villars

On p. 8 each propagator is multiplied by $\Lambda^2/(p^2+\Lambda^2)$, which makes it a particle of mass m minus a negative-norm ghost of mass Λ, both of weight $\Lambda^2/(\Lambda^2-m^2)$ (§10.15, Eq. (10.15.10)). The loop integrand falls like $\bar{q}^{-8}$, as I'''s does: hence p. 8's "finite for $d<8$". With masses $M_a, M_b \in \{m, \Lambda\}$, signs $s_m = 1$, $s_\Lambda = -1$ and $D_{ab} = x(1-x)k^2 + xM_a^2 + (1-x)M_b^2$, the four $d=6$ integrals still diverge one by one. But $F(D) = \int d^6\bar{q}\,(2\pi)^{-6}(\bar{q}^2+D)^{-2}$ has $F''(D) = (4\pi)^{-3}/D$ (Route 1 again), so $F = (4\pi)^{-3}[D\ln D + a + bD]$, where a, b do not depend on D and diverge as a cutoff is removed. They cancel, since $\sum s_a s_b = 0$ and $\sum s_a s_b D_{ab} = 0$ at every x:

$$(4\pi)^3 I_{\text{PV}} = \Big(\frac{\Lambda^2}{\Lambda^2-m^2}\Big)^2 \sum_{a,b} s_a s_b \int_0^1 dx\, D_{ab}\ln D_{ab}. \tag{10.19.6}$$

For $\Lambda \gg m$,

$$(4\pi)^3 I_{\text{PV}} = \frac{\Lambda^2}{2} - \Big(\frac{k^2}{6}+m^2\Big)\ln\frac{\Lambda^2}{m^2} + \int_0^1 dx\, D\ln\frac{D}{m^2} + \frac{3m^2}{2} + \frac{k^2}{9} + O\Big(\frac{\ln\Lambda}{\Lambda^2}\Big). \tag{10.19.7}$$

There is a Λ^2 and a $\ln\Lambda^2$, both of the form $a+bk^2$, and no Λ^4 (p. 8's "$\propto\Lambda^4$" does not occur in $d=6$). Letting $\Lambda\to\infty$ cancels nothing; A and B do. Matching $\frac{1}{2}\alpha(4\pi)^3 I_{\rm PV} - Ak^2 - Bm^2$ to (10.19.3) gives the first row of the table below; Bm^2 carries $+\frac{1}{4}\alpha\Lambda^2$.

Route 3: dimensional regularization

In $d=6$, p. 6's condition fails: $(\bar q^2+D)^{-2}$ falls only like $\bar q^{-4}$, and both sides of $d^dq = i\,d^d\bar q$ diverge. The q^0 rotation itself is harmless at fixed q in any d: the integrand falls like q_0^{-4}, so the arcs vanish, and with the poles in quadrants II and IV (§10.8), $\int dq^0(-q_0^2+E^2-i\epsilon)^{-2} = i\pi/2E^3$ is i times the imaginary-axis integral ($E^2=\mathbf{q}^2+D$). What fails is the remaining $(d-1)$-dimensional integral of E^{-3}, which converges only for $d<4$, where it gives (10.18.1), $(4\pi)^{-d/2}\Gamma(2-\frac{d}{2})D^{d/2-2}$. This is analytic in d except for poles at $d=4,6,8,\ldots$; dim reg *defines* the $d=6-\epsilon$ integral as its continuation (p. 9). With $g\to g\tilde\mu^{\epsilon/2}$, $\mu^2 = 4\pi e^{-\gamma}\tilde\mu^2$ (pp. 10–11), $\Gamma(-1+\frac{\epsilon}{2}) = -\frac{2}{\epsilon}+\gamma-1+O(\epsilon)$ and $\int_0^1 D\,dx = \frac{k^2}{6}+m^2$,

$$(4\pi)^3\tilde\mu^\epsilon I = -\Big(\frac{2}{\epsilon}+1+\ln\frac{\mu^2}{m^2}\Big)\Big(\frac{k^2}{6}+m^2\Big) + \int_0^1 dx\,D\ln\frac{D}{m^2} + O(\epsilon). \qquad (10.19.8)$$

The $-(k^2/6+m^2)$ that multiplies $\ln(\Lambda^2/m^2)$ in (10.19.7) multiplies $2/\epsilon+\ln(\mu^2/m^2)$ here: $\ln\Lambda^2 \leftrightarrow 2/\epsilon$, as in $d=4$ (§10.18); the $\frac{1}{2}\Lambda^2$ has no counterpart. The pole is of the form $a+bk^2$ too, since $\partial_{k^2}^2[\Gamma(-1+\frac{\epsilon}{2})D^{1-\epsilon/2}] = \Gamma(1+\frac{\epsilon}{2})\,x^2(1-x)^2D^{-1-\epsilon/2}$ returns (10.19.4) at $\epsilon=0$. Matching to (10.19.3) gives p. 12's A and B, the second row of the table.

Same Π, different counterterms

	$-A/\alpha$	$-B/\alpha$
Pauli–Villars, on shell	$\frac{1}{6}\big[\ln\frac{\Lambda}{m} - \frac{1}{3} + \kappa_A\big]$	$\ln\frac{\Lambda}{m} - \frac{\Lambda^2}{4m^2} - \frac{3}{4} + \kappa_B$
dim reg, on shell	$\frac{1}{6}\big[\frac{1}{\epsilon} + \ln\frac{\mu}{m} + \frac{1}{2} + \kappa_A\big]$	$\frac{1}{\epsilon} + \ln\frac{\mu}{m} + \frac{1}{2} + \kappa_B$
dim reg, $\overline{\rm MS}$ (LecNote15)	$\frac{1}{6\epsilon}$	$\frac{1}{\epsilon}$

One-loop counterterms; Pauli–Villars up to $O(\Lambda^{-2}\ln\Lambda)$.

With the on-shell rows the three routes give one Π. At finite Λ the Pauli–Villars one (from (10.19.6) exactly, A and B fixed on shell) approaches it like m^2/Λ^2: at $k^2=2m^2$ it is 0.07531879, 0.07670498, 0.07671948 (units αm^2) for $\Lambda^2/m^2 = 10^2$, 10^4, 10^6, against 0.07671963 from (10.19.2). The counterterms differ, and not by a relabeling: $\ln\Lambda\to 1/\epsilon+\ln\mu+c$ turns the first row into the second only with $c=\frac{5}{6}$ for A but $c=\frac{5}{4}$ for B (and the Λ^2 dropped). $\ln\Lambda^2\leftrightarrow 2/\epsilon$ matches divergences, not finite parts. No paradox: A and B are not observables (§10.15 identifies $Z_\varphi-1$ with $-\int\rho$); Π, with m the pole mass, is. With the $\overline{\rm MS}$ row, $\Pi = \frac{1}{2}\alpha\int dx\,D\ln(D/\mu^2) - \frac{\alpha}{12}(k^2+6m^2)$

depends on μ and meets the two on-shell conditions at no μ: there m is not the pole mass (§10.24).

What each route hides. The regulator-free route hides the counterterms, and with them the $1/\epsilon$ poles that drive the running in $\overline{\text{MS}}$ (§10.24). Pauli–Villars shows the power divergence, $\frac{1}{4}\alpha\Lambda^2$ in Bm^2, the sensitivity of m^2 to a heavy scale behind the hierarchy problem (§10.18), at the price of a ghost: at finite Λ the residue is $\Lambda^2/(\Lambda^2 - m^2)$, not 1, and unitarity fails (§10.15). Dim reg adds no particle and needs one Gamma function, but the Λ^2 is gone.